MTH642 Quiz#1

Topic (1-44)

Due Date (23-25)Dec2021

- Mth642 quiz 1.Silent 25dec2021 pg20
- MTH642 Quiz 1 Solved 25dec2021 pg20
- Mth642 Quiz.1 by (KING) 25dec2021 pg20
- Mth642.quiz 1.moeez 25dec2021 pg20
- Mth642 Quiz.1.By (KING of KING) 25dec2021 pg30
- Mth642 quiz 1 25dec2021 pg10
- Mth642 quiz 1 Solved 25dec2021 pg10
- Camscanner 12-25-2021 pg40
- Mth642 Quiz#1 (1to41)(25dec2021) Made by Stylish pg20
- Mth642 quiz 1 from 3 groups 25dec2021 pg1263
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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 07:37:02 AM, 24 December 2021)

Which on of the following is **NOT** the unit of viscosity?

Select the correct option

○	$kgms^{-2}$ ✓
○	Nsm^{-2}
◉	poise
○	Pa.s

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 07:39:29 AM, 24 December 2021)

The flow of low-viscosity fluids at high velocities is typically _____.

Select the correct option

○	none of these
○	laminar
○	transitional
◉	turbulent

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 07:40:07 AM, 24 December 2021)

____ is the measure of internal thickness of the fluid.

Select the correct option

○	Stress
◉	Viscosity ✓
○	Volume
○	Momentum

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MTH642 - Fluid Mechanics (Quiz 1) **Quiz Start Time: 07:35 AM, 24 December 2021**

Question # 2 of 10 (Start time: 07:36:45 AM, 24 December 2021) **Total Marks: 1**

Fully developed velocity profile V=V(r) represents _____.

Select the correct option

Reload Math Equations

◉	one dimensional flow 
○	two dimensional flow
○	three dimensional flow
○	dimensionless flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 04:58:28 PM, 25 December 2021)

The fluid velocity _____ with depth.

Select the correct option

○	has no relation
○	increases ✓
○	decreases
○	remains constant

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 08:16:50 PM, 25 December 2021)

The dimension of area is _____

Select the correct option

- ○ L^{-2}
- ○ L^{2} ✓
- ○ L^{-1}
- ○ L^{3}

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Question # 4 of 10 (Start time: 04:48:51 PM, 25 December 2021)

Which property allows us to assume that the properties vary continuously in space without any jump discontinuities?

Select the correct option

- ○ Volume
- ○ Pressure
- ○ Continuum ✓
- ○ Density

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 1 of 10 (start time: 04:55:44 PM, 25 December 2021)

The magnitude of the drag force depends on _____.

Select the correct option

○	the velocity of the flow
○	the viscosity of the flow ✓
○	the pressure of the flow
○	the momentum of the flow

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Question # 10 of 10 (Start time: 04:00:32 PM, 25 December 2021)

At which vertical distance, the pressure in a fluid will be minimum while assuming that the fluid is at rest?

Select the correct option

○	10m
○	20m
○	7m ✓
○	9m

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MTH642 – Fluid Mechanics (Quiz 1)

Question # 8 of 10 (Start time: 04:40:14 PM, 25 December 2021)

Which of the following is **NOT** an internal flow?

Select the correct option

○	hot gas flow in a duct
○	unbounded fluid flow over the surface of a wire ✓
○	liquid flow between the two parallel plates
○	water flow in a pipe

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 04:41:29 PM, 25 December 2021)

The term **transient** is used for _____.

Select the correct option

○	steady flow
○	developing flow ✓
○	unsteady flow
○	any flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 03:38:31 PM, 25 December 2021)

If at any point the fluid properties change with respect to time, then the flow is termed as _____.

Select the correct option

○	unsteady flow ✓
○	uniform flow
○	steady flow
○	nonuniform flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 03:56:42 PM, 25 December 2021)

The quantity of matter or the region in space chosen for study is defined as _____

Select the correct option

○	boundary
○	system ✓
○	imaginary surface
○	surrounding

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MTH642 - Fluid Mechanics (Quiz 1)

Quiz Start Time: 03:53 PM, 25 December 2021

Question # 6 of 10 (Start time: 03:57:50 PM, 25 December 2021)

Total Marks: 1

Fluid flows between two parallel plates. Assume that the lower plate is at rest while upper one is moving with a constant velocity *V*. The velocity of the fluid in contact with the lower plate is _____.

Select the correct option

Reload Math Equations

○	*2V*
○	*V/2*
○	zero 
○	*V*

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 8 of 10 (Start time: 03:37:39 PM, 25 December 2021)

A closed system can also be referred as _____.

Select the correct option

○	isolated system
○	control volume
○	control mass ✓
○	international system

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 2 of 10 (Start time: 08:00:31 PM, 25 December 2021)

The no slip condition happens due to _____

Select the correct option

○	pressure
○	density
○	velocity ✓
○	viscosity

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 04:58:26 PM, 25 December 2021)

Surface tension is defined as _____.

Select the correct option

○	force per unit length ✓
○	momentum per unit length
○	viscosity per unit length
○	stress per unit length

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Question # 6 of 10 (start time: 05:11:50 PM, 25 December 2021)

At which vertical distance, the pressure in a fluid will be maximum while assuming that the fluid is at rest?

Select the correct option

- 7m
- 9m
- 8m
- 6m

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 2 of 10 (Start time: 07:59:55 PM, 25 December 2021)

Stress is defined as _____.

Select the correct option

○	force per unit area ✓
○	momentum per unit area
○	work done per unit area
○	fluid flow per unit area

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MTH642 - Fluid Mechanics (Quiz1)

Question # 2 of 10 (Start time: 06:35:40 PM, 25 December 2021)

The unit of velocity is _____.

Select the correct option

○	ms
○	$m^{-1}s$
○	$m^{-1}s^{-1}$
○	ms^{-1} ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 07:13:09 PM, 25 December 2021)

_____ is the secondary dimension.

Select the correct option

- Energy ✓
- Temperature
- Length
- Time

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 09:55:02 PM, 25 December 2021)

If the fluid is at rest, then which of the following will be zero?

Select the correct option

◉	shear stress 
○	pressure
○	normal stress
○	stress

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 8 of 10 (Start time: 05:23:53 PM, 25 December 2021)

In steady flow, _____

Select the correct option

○	properties gradually change with time.
○	velocity does not change
○	pressure changes
○	properties do not change with time ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 05:26:08 PM, 25 December 2021)

In a steady flow, the fluid properties _____ at any fixed point.

Select the correct option

○	vary
○	remain constant

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 09:24:55 PM, 25 December 2021)

Surface tension comes into act due to the _____ between the molecules of the liquids.

Select the correct option

○	spaces
○	attractive forces ✓
○	velocity difference
○	energy difference

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 8 of 10 (Start time: 05:01:05 PM, 25 December 2021)

Which of the following is a Newtonian fluid?

Select the correct option

○	Liquid plastics
○	Gasoline ✓
○	Blood
○	Toothpaste

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 03:33:01 PM, 25 December 2021)

The acceleration of the fluid particle is given by_____.

Select the correct option

○	$\mathbf{a}_{particle}=d\mathbf{P}_{particle}/dt$
○	$\mathbf{a}_{particle}=d\mathbf{T}_{particle}/dt$
○	$\mathbf{a}_{particle}=d\mathbf{V}_{particle}/dt$ ✓
○	$\mathbf{a}_{particle}=d\mathbf{x}_{particle}/dt$

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 10:57:37 PM, 25 December 2021)

An open system can also be referred as _____.

Select the correct option

○	international system
○	control mass
◉	control volume 
○	isolated system

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 10:58:00 PM, 25 December 2021)

Can a body weighing 150 lbf on earth will weigh only 25 lbf on the moon?

Select the correct option

◉	Yes 
○	No

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Question # 9 of 10 (Start time: 10:58:50 PM, 25 December 2021)

By Newton's second law, the net force acting on the fluid particle is given as the product of its_____ and_____.

Select the correct option

○	pressure, velocity
○	velocity, time
○	acceleration, time
◉	mass, acceleration 

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MTH642 - Fluid Mechanics (Quiz 1) Quiz Start Time: 10:56 PM, 25 December 2021

Question # 2 of 10 (Start time: 10:57:00 PM, 25 December 2021) Total Marks: 1

The _____ is commonly used to replace pumps by placing the water tank sufficiently above the solar collectors.

Select the correct option

Reload Math Equations

○	Compton effect
○	thermal effect
○	buoyancy effect
○	thermosiphoning effect

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 10:57:10 PM, 25 December 2021)

The friction between the fluid layers causes the fluid velocity to

Select the correct option

◉	decrease
○	increase

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Question # 4 of 10 (Start time: 10:57:24 PM, 25 December 2021)

The three terms **acceleration of a fluid particle, local acceleration and advective acceleration** can be related as

Select the correct option

○	acceleration of a fluid particle = local acceleration - advective acceleration
○	acceleration of a fluid particle = local acceleration × advective acceleration
○	acceleration of a fluid particle = local acceleration / advective acceleration
◉	acceleration of a fluid particle = local acceleration + advective acceleration ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 10:48:48 PM, 25 December 2021)

Which of the following is called **the local acceleration**?

Select the correct option

○	dV/dt
○	DV/Dt
○	$\Delta V/\Delta t$
◉	$\partial V/\partial t$

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 10:50:11 PM, 25 December 2021)

Can a body weighing 150 lbf on earth will weigh only 25 lbf on the moon?

Select the correct option

○	No
○	Yes 

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 10:50:58 PM, 25 December 2021)

In _____, each fluid particle follows the identical path.

Select the correct option

○	fluid flow
○	laminar flow
◉	turbulent flow
○	developing flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 10:47:56 PM, 25 December 2021)

If a fluid is at rest, then shear stress equals to_____.

Select the correct option

◉	zero 
○	none of these
○	pressure
○	normal stress

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 10:47:04 PM, 25 December 2021)

Viscosity is caused by _____.

Select the correct option

○	cohesive forces between the molecules in liquids
○	friction force develops between two fluid layers when they move
○	All ✓
○	molecular collisions in gases

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 10:40:36 PM, 25 December 2021)

Which of the following statement(s) is/are true?

1. A liquid does not form a free surface.
2. Gas expands to fill the entire available space.

Select the correct option

◉	1 and 2 both ✓
○	Neither 1 nor 2
○	2 only
○	1 only

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 10:42:40 PM, 25 December 2021)

The local acceleration will be zero for

Select the correct option

○	unsteady flow
○	steady flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 1 of 10 (Start time: 10:45:44 PM, 25 December 2021)

Flow having significant frictional effects is termed as _____.

Select the correct option

◉	viscous flow ✓
○	inviscid flow
○	external flow
○	internal flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 10:34:50 PM, 25 December 2021)

The flow is _____ for Mach number less than one.

Select the correct option

◉	subsonic
○	supersonic
○	sonic
○	hypersonic.

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 10:35:43 PM, 25 December 2021)

The are _____ primary dimensions.

Select the correct option

○	six
○	seven 
○	five
○	eight

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Question # 2 of 10 (Start time: 10:40:05 PM, 25 December 2021)

The Greek mathematician Archimedes applied _____ to determine the gold content of the crown of King Hiero.

Select the correct option

○	the law of conservation of mass
◉	the buoyancy principle
○	the gravity principle
○	the law of conservation of momentum

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 10:30:26 PM, 25 December 2021)

By definition, the acceleration of the fluid particle is the time derivative of __

Select the correct option

○	the fluid's pressure
◉	the fluid's velocity 
○	the fluid's momentum
○	the fluid's displacement

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 10:32:18 PM, 25 December 2021)

The advective acceleration is given by _____.

Select the correct option

○	$V \bullet \nabla$
○	$\nabla \bullet V$
◉	$(V \bullet \nabla)V$
○	$\partial V / \partial t$

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 1 of 10 (Start time: 10:33:58 PM, 25 December 2021)

For fluid at rest, the shear stress is

Select the correct option

○	undefined
○	maximum
◉	zero 
○	minimum

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 10:28:57 PM, 25 December 2021)

The number of significant digits in 1.500 is

Select the correct option

○	3
○	1
◉	4 
○	2

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 10:29:43 PM, 25 December 2021)

In which direction, the pressure will not change if the fluid is at rest?

Select the correct option

○	direction along the negative slop
○	vertical direction
○	direction along the positive slop
○	horizontal direction ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 10:21:08 PM, 25 December 2021)

The substance in liquid or gas phase is referred as

Select the correct option

○	ions
◉	fluid 
○	electrolyte
○	plasma

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 10:21:44 PM, 25 December 2021)

Which of the following is **NOT** a drived dimension?

Select the correct option

○	Velocity
○	Length 
○	Volume
◉	Area

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MTH642 – Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 10:24:35 PM, 25 December 2021)

Which of the following statements is/are true?
1: Solids can resist applied shear stress by deforming.
2: liquids cannot resist applied shear stress and continuously deform.

Select the correct option

○	2 only
○	nether 1 nor 2
◉	both 1 and 2 ✓
○	1 only

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 10:28:16 PM, 25 December 2021)

In inviscid flow region, the viscous forces are negligibly small compared to____.

Select the correct option

○	Both (a) and (c)
◉	inertial or pressure forces ✓
○	gravitational forces
○	buoyant forces

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MTH642 - Fluid Mechanics (Quiz 1) **Quiz Start Time: 10:10 PM**

Question # 8 of 10 (Start time: 10:16:32 PM, 25 December 2021)

The pressure applied to a confined fluid increases the pressure throughout by the same amount. This law is known as _____

Select the correct option

Reloc

○	Euler's law
○	Pascal's law 
◉	Newton's law
○	Ohm's Law

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MTH642 – Fluid Mechanics (Quiz 1)

Question # 1 of 10 (Start time: 10:20:45 PM, 25 December 2021)

The flow regions in which the frictional effects are negligible are termed as _____ flow regions.

Select the correct option

○	external
◉	inviscid
○	viscous
○	internal

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MTH642 – Fluid Mechanics (Quiz 1)

Question # 2 of 10 (Start time: 10:20:56 PM, 25 December 2021)

The boundary layer is developed very close to the wall by flowing fluid due to _____.

Select the correct option

○	Gravity
◉	Viscosity ✓
○	Density
○	Surface Tension

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Question # 2 of 10 (Start time: 10:12:03 PM, 25 December 2021)

In what type of flow, the mass, volume and energy content of the flow section do not change?

Select the correct option

○	Steady flow ✓
○	Uniform flow
○	Nonuniform flow
○	Unsteady flow

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Question # 5 of 10 (Start time: 10:13:54 PM, 25 December 2021)

A fluid flow is classified as viscous or inviscid flow region, depending on_____.

Select the correct option

○	internal stress of the fliuid
○	fluid momentum
◉	frictional effects between the layers of the fluid ✓
○	internal temperature of the fluid

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Question # 6 of 10 (Start time: 10:14:41 PM, 25 December 2021)

In which of the following system, the amount of mass is fixed and no mass and even energy can cross its boundary?

Select the correct option

○	International system
○	Open system
○	Closed system 
○	Isolated system

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 10:07:04 PM, 25 December 2021)

A flow may be approximated _____ when the aspect ratio is large.

Select the correct option

○	three-dimensional
○	one-dimensional
○	zero-dimensional
○	two-dimensional 

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 10:07:44 PM, 25 December 2021)

If a normal force of one newton is exerted by the fluid per unit area, then the pressure will be _____.

Select the correct option

○	1 atm
○	1 bar
○	1kgf /cm^2
◉	1 Pa ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 1 of 10 (Start time: 10:10:50 PM, 25 December 2021)

In which of the following, the rate of deformation is proportional to the shear stress?

Select the correct option

◉	Newtonian fluids ✓
○	Dilatant
○	Pseudoplastics
○	Bingham Plastics

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MTH642 – Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 09:56:58 PM, 25 December 2021)

Barometer is a device used to measure _____.

Select the correct option

◉	atmospheric pressure 
○	partial pressure
○	saturation pressure
○	vapor pressure

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 10 of 10 (Start time: 09:57:15 PM, 25 December 2021)

If there is no change with location over a specified region then the flow is termed as _____.

Select the correct option

◉	steady flow
○	unsteady flow
○	nonuniform flow
○	uniform flow 

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Question # 3 of 10 (Start time: 10:04:09 PM, 25 December 2021)

The pressure exerted by the vapor of a pure substance in phase equilibrium with its liquid at given temperature is called __

Select the correct option

Rel

○	vapor pressure 
○	saturation pressure
○	partial pressure
◉	atmospheric pressure

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 10:04:26 PM, 25 December 2021)

The number of significant digits in 10.660 is _____.

Select the correct option

○	2
○	4
◉	5
○	3

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 7 of 10 (Start time: 09:55:44 PM, 25 December 2021)

The mathematical form of fluid properties, velocity (V) and density (ρ), in unsteady flow is _

Select the correct option

◉	$\frac{\partial V}{\partial t} \neq 0 \; and \; \frac{\partial \rho}{\partial t} \neq 0$
○	$\frac{\partial V}{\partial t} \neq 0 \; and \; \frac{\partial \rho}{\partial t} = 0$
○	$\frac{\partial V}{\partial t} = 0 = \frac{\partial \rho}{\partial t}$
○	$\frac{\partial V}{\partial t} = 0 \; and \; \frac{\partial \rho}{\partial t} \neq 0$

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 8 of 10 (Start time: 09:56:04 PM, 25 December 2021)

Manometer is a device used to measure _____.

Select the correct option

○	volume differences
○	velocity differences
○	pressure differences ✓
○	energy differences

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 09:52:48 PM, 25 December 2021)

If density is defined as mass per unit volume then what will be the specific volume?

Select the correct option

○	volume × unit mass
○	volume + unit mass
○	volume - mass
◉	volume / unit mass

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 4 of 10 (Start time: 09:53:04 PM, 25 December 2021)

Which one of the following is **NOT** a secondary dimension?

Select the correct option

◉	Temperature ✓
○	Acceleration
○	Momentum
○	Velocity

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 09:55:28 PM, 25 December 2021)

One-, Two- and Three-Dimensional flows are characterized by _____.

Select the correct option

○	energy distribution
○	mass distribution
◉	velocity distribution ✓
○	momentum distribution

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MTH642 - Fluid Mechanics (Quiz 1) **Quiz Start Time: 09:39 PM, 25 December 2021**

Question # 10 of 10 (start time: 09:47:41 PM, 25 December 2021) **Total Marks: 1**

Fluid flows between two parallel plates. Assume that the lower plate is at rest while upper one is moving with a constant velocity *V*. According to no-slip condition, the relative velocity of the fluid in contact with the upper plate is _____.

Select the correct option

Reload Math Equations

○	*2V*
○	zero
○	*V/2*
○	*V* ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 1 of 10 (Start time: 09:51:12 PM, 25 December 2021)

The fluid forced to flow in a confined channel is classified as being _____.

Select the correct option

○	natural flow
◉	viscous flow
○	internal flow 
○	compressible flow

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 2 of 10 (Start time: 09:51:35 PM, 25 December 2021)

Viscosity is a measure of internal _____ of the fluid.

Select the correct option

◉	thickness ✓
○	stress
○	momentum
○	temperature

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 09:40:53 PM, 25 December 2021)

Which of the following may **NOT** have a free surface?

Select the correct option

○	An open container filled with water
○	An open container filled with hydrogen gas ✓
○	An open container filled with oil
○	An open container filled with liquid mercury

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 09:43:28 PM, 25 December 2021)

The developing velocity profile **V(r,z)** represents _____.

Select the correct option

○	one dimensional flow
○	three dimensional flow
○	dimensionless flow
◉	two dimensional flow ✓

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 09:46:55 PM, 25 December 2021)

Which one of the following is **NOT** a primary dimension?

Select the correct option

○	Time
○	Area 
○	Length
○	Mass

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 2 of 10 (Start time: 09:40:05 PM, 25 December 2021)

For unsteady flows the local acceleration will be

Select the correct option

○	zero
◉	nonzero

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 10:48:32 PM, 25 December 2021)

Which of the following is an extensive property?

Select the correct option

○	Density
○	Pressure
◉	Total mass
○	Temperature

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 10:48:17 PM, 25 December 2021)

In solids stress is proportional to strain but in fluids stress is proportional to strain rate.

Select the correct option

○	False
◉	True



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MTH642 - Fluid Mechanics (Quiz 1)

Question # 9 of 10 (Start time: 08:06:18 PM, 25 December 202

Which one of the following is a primary dimension?

Select the correct option

○	Energy
○	Time 
○	Velocity
○	Volume

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 6 of 10 (Start time: 06:39:57 PM, 25 December 2021)

The flow having velocity $\mathbf{V} = (-y, x)$ is _____.

Select the correct option

○	three-dimensional
○	one-dimensional
○	zero-dimensional
○	two-dimensional ✓

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MTH642 - Fluid Mechanics (Quiz1)

Question # 3 of10 (Start time: 06:37:02 PM, 25 December 2021)

Newton's second law applied to the fluid particle, is given as _____.

Select the correct option

○	$\mathbf{F}_{particle} = m_{particle} - \mathbf{a}_{particle}$
○	$\mathbf{F}_{particle} = m_{particle} * \mathbf{a}_{particle}$ ✓
○	$\mathbf{F}_{particle} = m_{particle} / \mathbf{a}_{particle}$
○	$\mathbf{F}_{particle} = m_{particle} + \mathbf{a}_{particle}$

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Question # 10 of 10 (Start time: 05:13:05 PM, 25 December 2021)

At which vertical distance, the pressure in a fluid will be minimum while assuming that the fluid is at rest?

Select the correct option

- ○ 20m
- ○ 9m
- ○ 7m
- ○ 10m

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 3 of 10 (Start time: 04:57:20 PM, 25 December 2021)

The developing flow in a circular pipe, in cylindrical coordinate system, is _____

Select the correct option

- ○ one-dimensional
- ○ zero-dimensional
- ○ two-dimensional ✓
- ○ three-dimensional

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MTH642 - Fluid Mechanics (Quiz 1)

Question # 5 of 10 (Start time: 03:56:42 PM, 25 December 2021)

The quantity of matter or the region in space chosen for study is defined as _____.

Select the correct option

○	boundary
○	system ✓
○	imaginary surface
○	surrounding

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